

Curtin University – Department of Computing

Assignment Cover Sheet / Declaration of Originality

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Last name:	Shrenik Jobanputra	Student ID:	19128014
Other name(s):			
Unit name:	Machine Perception	Unit ID:	COMP 3007
Lecturer / unit coordinator:	Senjian An	Tutor:	Sam
Date of submission:	5/10/2020	Which assignment?	1 (Leave blank if the unit has only one assignment.)

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- It is my responsibility to ensure that my submission is complete, correct and not corrupted.

Signature: Shrenik Jobanputra	Date of signature: 5/10/2020
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Machine Perception Assignment 1

By : Shrenik Jobanputra

Student Number : 19128014

I. INTRODUCTION

Assignment 1 consisted us to perform 4 tasks involving the 2 images : Diamond and Dugong. Task 1, 3 and 4 were completed but Task 2 was not fully completed.

All the codes for the Tasks are under the Appendices.

II. TASK 1

A. Question 1

Image Histograms represent the Intensity of Pixels VS the Number of Pixels at the specific intensity. For images that are rotated, the Image Histogram will remain the same as is for the original image. This is because the actual Intensity of the pixels DO NOT CHANGE! And, the number of pixels at a specific intensity DO NOT CHANGE either, hence Image Histogram is INVARIANT for Rotation of Image.

However, for Scaling, Image Histograms do change in terms of their Y-Axis, which is the number of pixels at a specific Intensity. For example, doubling the dimensions of an image produces 4 times as many pixels as the original image. As the actual intensities stay the same, only the number of pixels for a given intensity will change on the Image Histogram. Hence Image Histogram is VARIANT for Scaling.

Harris Corner is Rotation INVARIANT as if the image is rotated, we can find the same corners. This is obvious because the corners remain as corners in a rotated image.

However Harris Corner is Scale VARIANT, because when the size of an image is increased, the corner in the larger image appears more flatter.

SIFT (Scale Invariant Feature Transform) is both Rotation invariant and Scale invariant, as it gets multiple Gaussian Blurred Images, finds the difference between 2 of them and marks them. It is still rotation invariant as the features remain the same even when the image is rotated.

B. Question 2

- ATTEMPTED WITH CODE -

C. Question 3

The claims in Question 1 of Task 1 can be verified with the experimental data as will be shown in the following figures 1,2,3,4 and 5.

Some difficulties that were encountered initially were that the Image Histogram for the Diamond Image had a big 'Shoot' on intensity of 255 with the most number of pixels. This is because most of the Diamond image is white, and white has an intensity of 255 for all colours. This 'shoot' created the other intensities to not be visible as much in the Image Histogram as the Number of Pixels of intensities lowers than 255 is minuscule compared to the Number of Pixels for intensity of 255.

To solve this issue mentioned above, the histogram range was then reduced to 250. So only from 0 – 250 intensity of pixels are then plotted on the histogram, which makes the other intensities stand out more.

Another issue I had was regarding rotation of the images. Rotation from OpenCV caused the images to be slightly cropped after the rotation. Hence to solve this, I increased the Background size (which is Black in colour), and this encompassed the entire rotated image. However another issue arose when doing the Image Histogram of this, as now most of the pixels had an intensity of 0 (because black has intensity of 0) !!!

To solve this issue mentioned above, the histogram minimum to 10. So only from 10 – 250 (from the previous issue) intensity of pixels are then plotted on the histogram, which makes the other intensities stand out more.

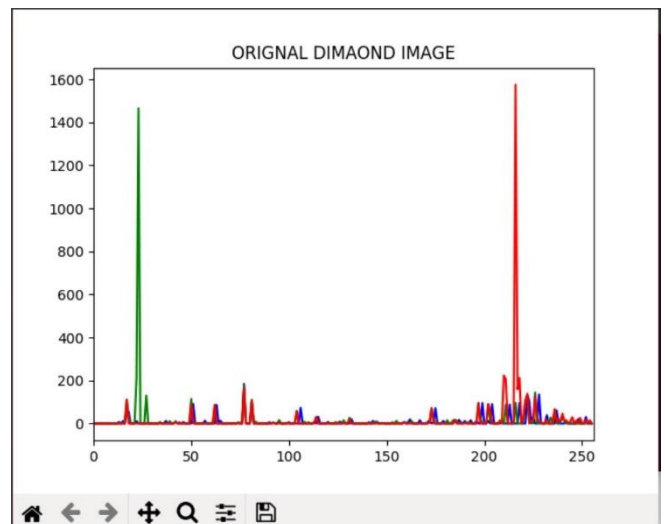


Figure 1: Histogram of the Original Diamond Image

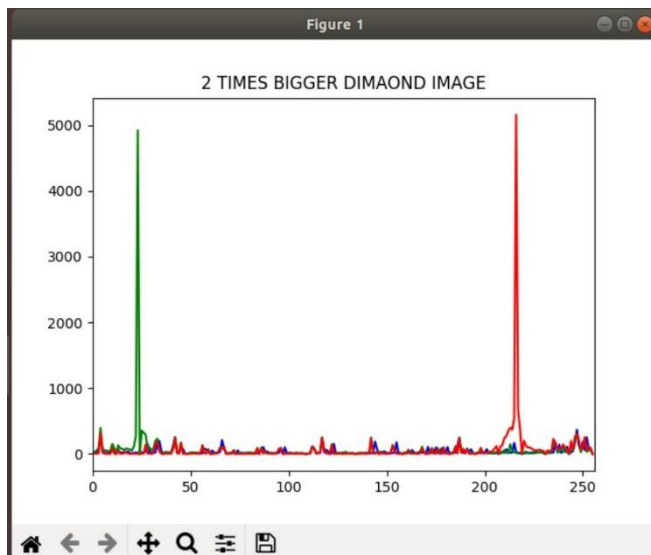


Figure 2: Histogram of the Diamond image doubles in size with all dimensions

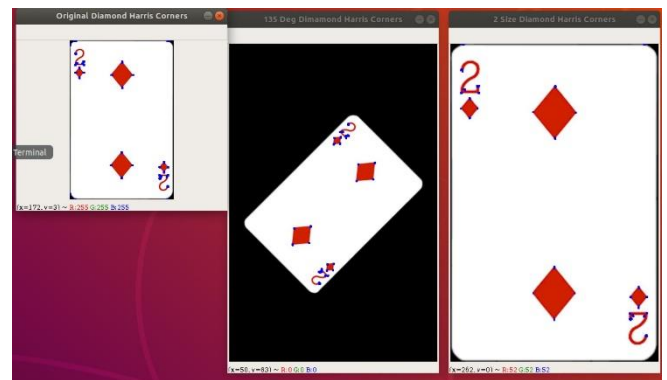


Figure 4: Comparisons of Harris Corners between Diamond Original, Double size and 135 degree Anticlockwise rotated.

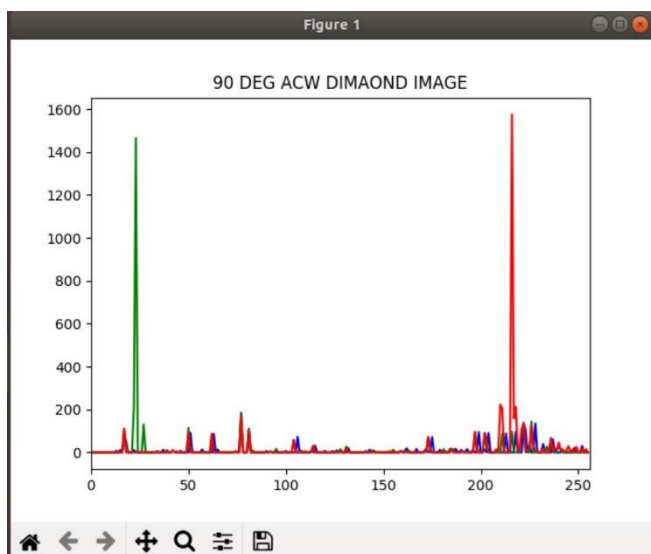


Figure 3: Histogram of the Diamond Image rotated 90 degree Anticlockwise

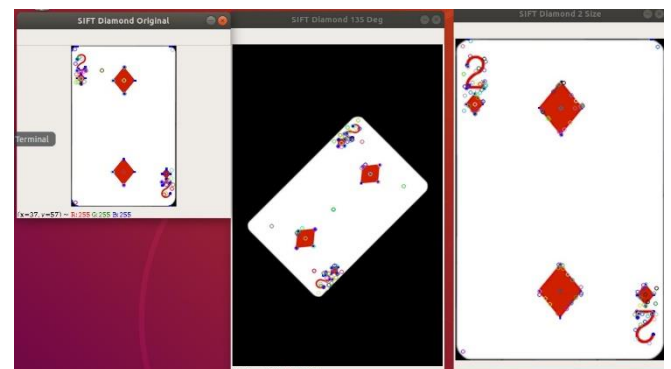


Figure 5: Comparisons of SIFT between Diamond Original, Double size and 135 degree Anticlockwise rotated.

III. TASK 2

A. Question 1

The main steps of extractions of SIFT are : keypoint Detection, keypoint localization, orientation assignment and then feature descriptor generation. SIFT gets multiple Gaussian Blurred Images, finds the difference between the 2 of them and marks them.

The main steps for LBP(Local Binary Patterns) is the window is divided into cells, then, each cell has 16 x 16 pixels. Then, we compare the value of each pixel with 8 neighbour following a specific order. Then we encode 1/0 if the pixel value is less/larger than the neighbour pixel. Then we output this 8-bit binary number.

The main steps of HOG(Histogram Of Gradients) is the preprocessing, the calculating the Gradient Images using Gaussian Derivatives. Then calculating the HOG in 8x8 cells. Then we normalize the block, and then we calculate the HOG feature vector!

SIFT is more suitable for images that are from different views such as a 3D Scene Reconstruction. Whereas HOG is better for images with lots of imperfections or noise, as this gives all the gradients and allows us to match similar gradients.

B. Question 2

- ATTEMPTED WITH CODE -

C. Question 3

This cannot be completed as the images from Task 2 for HOG and SIFT were not analyzed. Although the steps for the analysis would be as follows:

1. Get a Key Point on the original image
2. Get a Feature Vector for the Key point of original image
3. Now on a changed image(rotated or scaled), get the same Key Point and find its feature vector
4. Now find the distance between the two feature vectors and compare it with the normalized
5. If the distance is big compared to the normalized, then it is the method is variant, else it is invariant!

IV. TASK 3

A. Question 1

ATTEMPTED WITH CODE -

B. Question 2

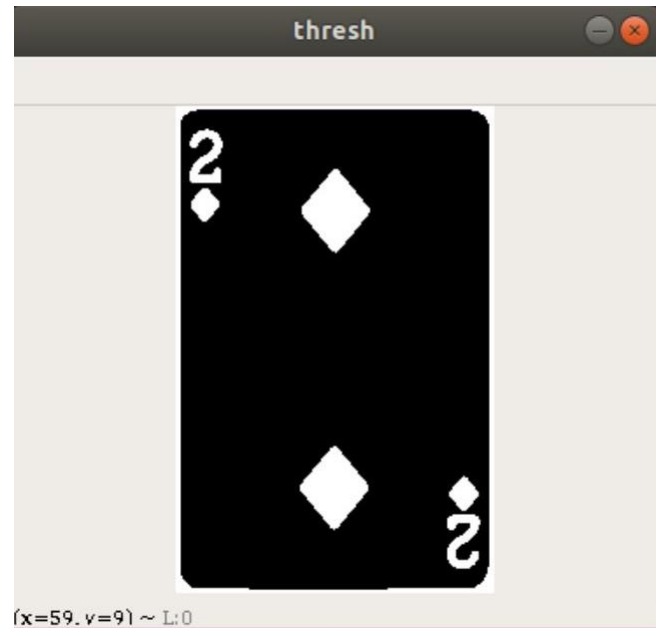


Figure 6: Binary Image of Diamond using thresholding

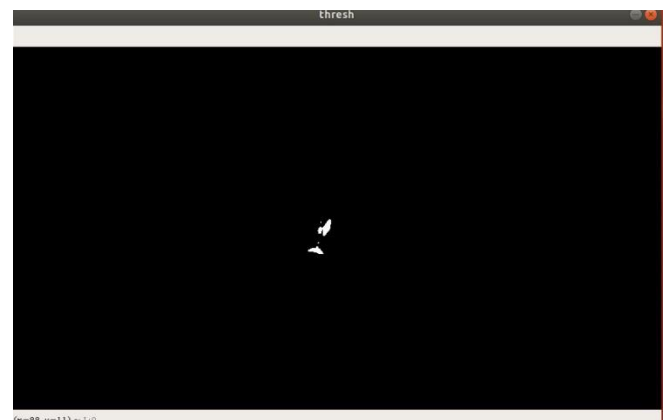


Figure 7: Binary Image of Dugong using thresholding

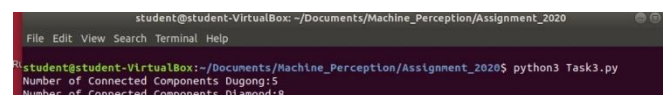


Figure 8: Number of Connected Components for the Diamond and the Dugong Image

v. TASK 4

The Image Segmentation was conducted with K-means for each of the images. For first method I used a combination of ALL the channels. The second method was using a combination of Blue and Red Channel ONLY! And the third method was using Green and Red Channel ONLY and then performing K-means on each of this. The Figures below show the results.

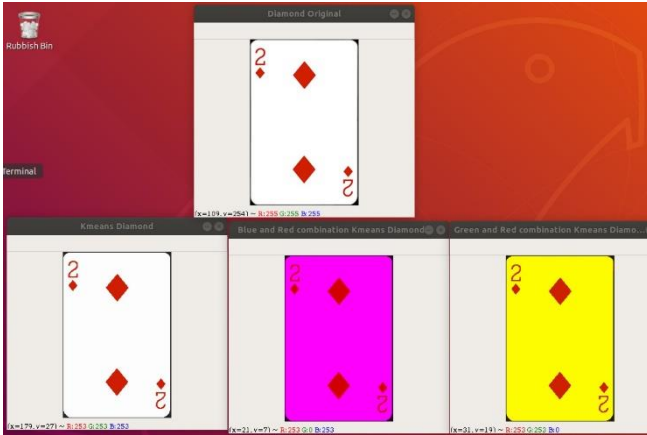


Figure 9: Comparisons of the K-means with different combination of channels

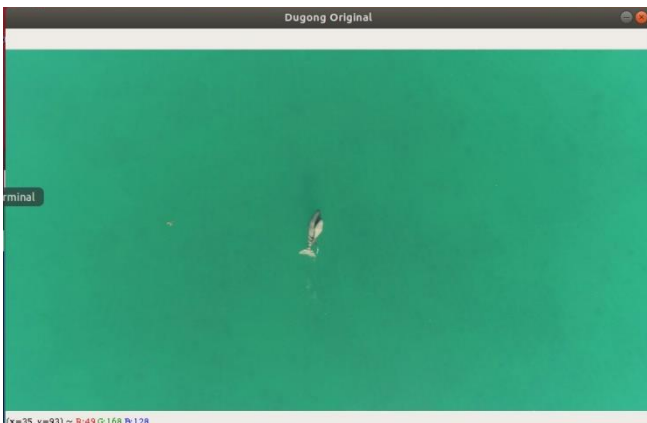


Figure 10: Original Dugong Image



Figure 11: Kmeans of Dugong Image with all colours



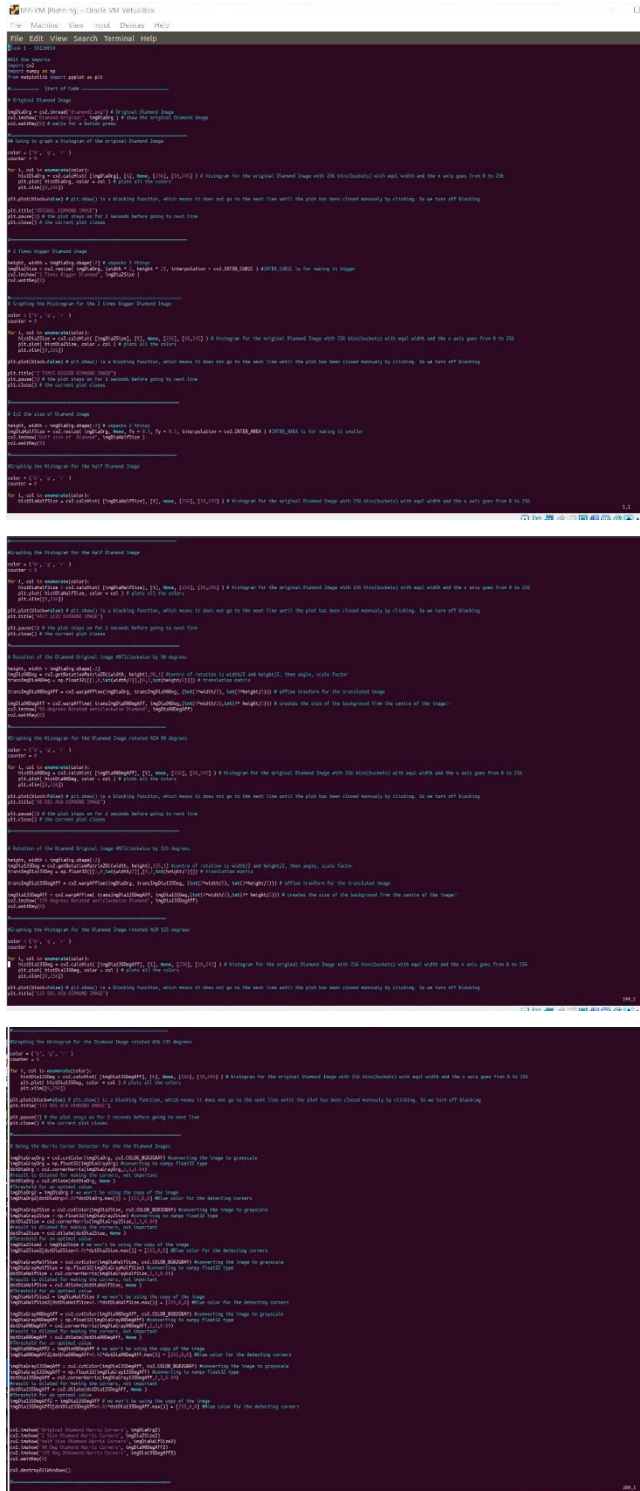
Figure 12: Kmeans of Dugong Image with Blue and Red colours ONLY



Figure 13: Kmeans of Dugong Image with Green and Red colours ONLY

APPENDIX 1

Code for Task 1

[illegible][illegible][illegible]

APPENDIX 2

Code for Task 2

[illegible][illegible][illegible][illegible]

APPENDIX 3

Code for Task 3

[illegible]

APPENDIX 4

Code for Task 4

[illegible]

References

- Subscription.packtpub.com. 2020. *{{Metadatacontroller.Pagetitle}}*. [online] Available at: <https://subscription.packtpub.com/book/application_development/9781788396905/1/ch01/v1/sec15/image-rotation> [Accessed 5 October 2020].
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